PRE-ALGEBRA (8TH) · REAL NUMBERS AND EXPONENTS

Square Roots on the Number Line

Explore & Reflect **TEACHER KEY**

Name _____ Date _____ Period _____

Open **Square Roots on the Number Line** in your unit's Interactive Math section on Mr. K's Classroom (or use your teacher's link). Answer Part 1 **before** you use the tool. Use the tool for Parts 2 and 3. Do Parts 4 and 5 on your own.

1 Predict before you touch the tool

1. Between which two whole numbers is √40? How do you know without a calculator?
   6 and 7, because $6^2 = 36 < 40 < 49 = 7^2$.
2. Is √50 equal to 25? Explain.
   No: $25^2 = 625$, not 50. √50 ≈ 7.07.

2 Explore use the tool

1. Choose area 40 and zoom in once. Between which tenths is √40?
   6.3 and 6.4 ($6.3^2 = 39.69$, $6.4^2 = 40.96$).
2. Zoom in again. Between which hundredths is √40?
   6.32 and 6.33 ($6.32^2 = 39.9424$, $6.33^2 = 40.0689$).
3. Choose area 49 and zoom in. What is different?
   √49 = 7 exactly. It is a perfect square, so zooming finds nothing new.
4. Type estimates for √20 until the square is within 0.05 of 20. Record your best estimate and its square.
   4.47, which squares to 19.9809 (answers may vary slightly).

3 Record use the tool and fill in the table

Trap the square root

Area	Between whole numbers	Between tenths	Rational or irrational?
30	5 and 6	5.4 and 5.5	irrational
64	8 exactly	—	rational
10	3 and 4	3.1 and 3.2	irrational
75	8 and 9	8.6 and 8.7	irrational
2	1 and 2	1.4 and 1.5	irrational

4 Explain what did you discover?

1. √A is the side length of a square whose area is A.
2. To trap √A, find two numbers whose squares are just below and just above A.
3. The square root of a whole number that is not a perfect square is irrational: its decimal never ends and never repeats.
4. Zooming showed √2 ≈ 1.414…. Does zooming prove that √2 is irrational? Explain.
   No. Zooming only shows more digits. It cannot show that the decimal never ends or repeats; that needs a mathematical proof.

5 Apply a new situation, no tool

1. Order from least to greatest: $\sqrt{30}$, 5.4, $\frac{11}{2}$, $\sqrt{36}$.

   $5.4 < \sqrt{30}$ (≈ 5.48) $< \frac{11}{2}$ (5.5) $< \sqrt{36}$ (6)

2. Estimate $\sqrt{90}$ to the nearest tenth without a calculator.

   9.5 ($9.4^2 = 88.36$ and $9.5^2 = 90.25$; 90 is closer to 90.25).

3. A square garden has an area of 200 ft². About how long is each side?

   $\sqrt{200} \approx 14.1$ ft ($14^2 = 196$, $14.2^2 = 201.64$).

4. Which of these are irrational: $\sqrt{16}$, $\sqrt{17}$, $\frac{1}{3}$, $\sqrt{0.25}$?

   Only $\sqrt{17}$.

Teacher notes

Intended discovery: $\sqrt{A}$ is a side length; squaring neighbors traps it one place value at a time. Perfect squares give exact whole numbers; other square roots of whole numbers are irrational, so every decimal is an approximation.

Likely misconceptions:

- $\sqrt{A} = A \div 2$.
- Believing a long calculator decimal is exact.
- Thinking zooming or a calculator proves irrationality.

Answers that can vary: Explore 4 estimates (any 2-decimal value whose square is within 0.05 of 20 is correct: only 4.47).